METHYLDOPA

Pregnancy: safe & first-line

Positive Coombs test

CHEMICAL NAME (2S)-2-amino-3-(3,4-dihydroxyphenyl)-2-methylpropanoic acid (sesquihydrate)

CLASS Centrally acting α2-adrenergic agonist antihypertensive

TRADE NAMES Aldomet · Dopegyt · Methyldopa (generic)

THERAPEUTIC USES / INDICATIONS

- Hypertension in pregnancy (long safety record).

PHARMACODYNAMICS

- Prodrug → **α-methylnoradrenaline** stimulates central **α2 receptors** → ↓ sympathetic outflow → ↓ peripheral resistance and BP.
- Does not reduce uteroplacental blood flow.

ADMINISTRATION

- **PO** 2–4 times daily.
- Increase dose gradually; evening dose increases help with sedation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Sedation, drowsiness, fatigue (often transient).
- Dry mouth, nasal congestion.
- Headache, depression.

ADVERSE EFFECTS (SERIOUS)

- Positive direct Coombs test (10–20%) and haemolytic anaemia (rare).
- Hepatitis.
- Drug fever, lupus-like syndrome.
- Postural hypotension; rebound hypertension on abrupt withdrawal.

DOSAGES

- **Adult:** 250 mg two to three times daily; increase every 2 days.
- Usual 500 mg–2 g/day in 2–4 divided doses; max 3 g/day.
- Elderly: start 125 mg twice daily.

CONTRAINDICATIONS & CAUTIONS

- Active liver disease; previous methyldopa hepatotoxicity.
- Depression (history).
- Phaeochromocytoma; acute porphyria.
- Use with MAOIs.

KEY INTERACTIONS

- **MAOIs:** hypertensive reaction.
- **Lithium:** ↑ toxicity.
- **Iron supplements:** ↓ absorption of methyldopa.
- **Anaesthetics:** ↑ hypotension.

PHARMACOKINETICS

- **Onset:** 3–6 h; maximum effect 6–8 h (full 2–3 days). **Duration:** 12–24 h.
- Converted in brain to α-methylnoradrenaline (active).
- Excreted in urine. **t½:** 1.5–2 h.

NURSING MANAGEMENT

- Monitor BP (including postural) and fetal wellbeing.
- Baseline and periodic CBC, Coombs, LFTs.
- Assess mood (postpartum depression risk; often switched after delivery).
- Inform blood bank (cross-matching difficulty).

PATIENT EDUCATION

- Drowsiness usually improves after a few days; avoid driving until it does.
- Do not stop suddenly.
- Report yellow skin, fever, low mood.
- Separate from iron tablets by 2 h.

KEY POINTS

- Central α2 agonist; first-line in pregnancy.
- Depression — switch after delivery.
- Sedation, positive Coombs, hepatitis.
- Do not stop abruptly.

SOURCES CHECKED **[342]** Methyldopa 125 mg Tablets (UK SmPC) **[311]** NICE NG133 **[343]** Methyldopa, oral (US label) **[344]** Methyldopa, oral (US label, Chartwell RX)

Full references p. 432–457